

Proposed Solution

Project Name: AI FAQ Assistant API

Project Design Phase | Date: 30/09/2026

Team ID: SWTID-2026-5919

Proposed Solution

S. No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Customers often face delays in getting answers to common questions. Manually managing FAQs and responding to repetitive queries requires time and effort. Finding relevant information in unorganized FAQ documents is also difficult.
2	Idea / Solution Description	The AI FAQ Assistant API is a backend application that automates FAQ management and customer support. It uses Node.js, Express.js, and MongoDB to store and manage FAQs. Google Gemini AI is integrated to generate questions and answers based on user-provided topics. The API also provides search functionality, user authentication, and secure FAQ management.
3	Novelty / Uniqueness	The system combines AI-powered FAQ generation with centralized FAQ storage and keyword-based search in a single API. It allows users to generate support content automatically, manage FAQ records, and retrieve relevant answers through REST API endpoints. Its modular design also allows integration with websites and mobile applications.
4	Social Impact / Customer Satisfaction	The proposed solution helps customers access information quickly and reduces the workload of customer support teams. It can improve customer satisfaction by providing convenient access to frequently requested information. It also helps organizations maintain consistent support content and reduce response delays.
5	Business Model (Revenue Model)	The application can follow a subscription-based model, offering basic FAQ management for free and advanced AI features through paid plans. Businesses can be charged based on the number of users, API requests, or AI-generated responses. Customized integration and enterprise support can also generate revenue.
6	Scalability of the Solution	The API follows a modular architecture that can be extended as the number of users and FAQ records increases. Cloud deployment, database indexing, caching, and load balancing can support higher traffic. Additional features such as multilingual support, semantic search, analytics, and integration with other customer support platforms can be introduced in the future.